

A Bohr–Bombieri Improvement of the Polynomial-Boundedness Constant

Automatic Math Discovery Run `fa_banach_001`

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Status

Status: **literature-implied answer (partial subcase)**. The source paper asks whether the constant $2\sqrt{3}/3$ in its displayed inequality (6.1) is optimal. The answer is no: the classical Bohr–Bombieri majorant formula lowers the constant to

$$B(1/2) = 6 - 2\sqrt{6} < \frac{2\sqrt{3}}{3}.$$

This settles the optimality question for that displayed estimate, but it does not answer the stronger question in the same paragraph asking for a norm-one operator that is polynomially bounded with constant 1 and has $\Psi(T) = \infty$.

Source Question

The source paper is H. Blazhko, D. Homza, F. L. Schwenninger, J. de Vries, and M. Wojtylak, *The algebraic numerical range as a spectral set in Banach algebras*, arXiv:2410.10678. In Example 6.3, after proving

$$\|f(T)\| = \max\{\|f(E)\|_2, \|f(\tfrac{1}{2}L)\|_p\} \leq \frac{2\sqrt{3}}{3} \sup_{\mathbb{D}} |f|,$$

the authors write:

$\|f(E)\|_2 \leq \sup_{\mathbb{D}} |f|$ for all polynomials f , while by Proposition 3.3 we have $\|f(\tfrac{1}{2}L)\|_p \leq \frac{2\sqrt{3}}{3} \sup_{\mathbb{D}} |f|$ for all polynomials f . We deduce that

$$(6.1) \quad \|f(T)\| = \max\{\|f(E)\|_2, \|f(\tfrac{1}{2}L)\|_p\} \leq \frac{2\sqrt{3}}{3} \sup_{\mathbb{D}} |f|$$

for all polynomials f .

Finally, we show that $\Psi(T, \mathcal{B}(X)) = \infty$. Since $V(\tfrac{1}{2}L, \mathcal{B}(\ell^p)) = \tfrac{1}{2}\overline{\mathbb{D}}$ and $\Psi(\tfrac{1}{2}L, \mathcal{B}(\ell^p)) = \infty$, there exists for each positive integer n some polynomial f_n such that

$$\|f_n(\tfrac{1}{2}L)\|_p \geq n \sup_{\tfrac{1}{2}\overline{\mathbb{D}}} |f_n|.$$

From this we deduce that

$$\|f_n(T)\| \geq \|f_n(\tfrac{1}{2}L)\|_p \geq n \sup_{\tfrac{1}{2}\overline{\mathbb{D}}} |f_n| = n \sup_{V(T, \mathcal{B}(X))} |f_n|$$

for all $n \in \mathbb{N}$ and the claim follows. It remains an open question whether the constant $\frac{2\sqrt{3}}{3}$ in (6.1) is optimal. In particular, it is unknown whether there exists an operator T of norm 1 which is polynomially bounded with constant 1 but with $\Psi(T) = \infty$.

We now turn our attention to combinatorial Banach spaces. We will show that for a large class of

Supporting Literature

Let $f(z) = \sum_{n \geq 0} a_n z^n$ be analytic on $\mathbb{D}$ with $\|f\|_{\infty, \mathbb{D}} \leq 1$. The Bohr–Bombieri function for the identity operator is

$$B(r) := \sup_{\|f\|_{\infty, \mathbb{D}} \leq 1} \sum_{n \geq 0} |a_n| r^n.$$

Bombieri and Ricci proved that, for $1/3 \leq r \leq 1/\sqrt{2}$,

$$B(r) = \frac{3 - \sqrt{8(1 - r^2)}}{r}.$$

This formula is recalled explicitly, with citation to Bombieri–Ricci, in Khasyanov’s 2025 survey/article on majorant series estimates [3, Section 4]. At $r = 1/2$ it gives

$$B(1/2) = 6 - 2\sqrt{6}.$$

Identification And Proof

Lemma 1. *Let L be the unilateral left shift on ℓ^p , $1 \leq p \leq \infty$. For every polynomial $f(z) = \sum_{k=0}^m a_k z^k$,*

$$\|f(\tfrac{1}{2}L)\|_p \leq (6 - 2\sqrt{6}) \sup_{|z| \leq 1} |f(z)|.$$

Proof. Since $\|L^k\|_p \leq 1$ for every k ,

$$\|f(\tfrac{1}{2}L)\|_p \leq \sum_{k=0}^m |a_k| 2^{-k}.$$

Applying the Bohr–Bombieri formula at $r = 1/2$, after normalizing by $\sup_{\mathbb{D}} |f|$, gives

$$\sum_{k=0}^m |a_k| 2^{-k} \leq (6 - 2\sqrt{6}) \sup_{\mathbb{D}} |f|.$$

□

Theorem 1. *In Example 6.3 of arXiv:2410.10678, the constant $2\sqrt{3}/3$ in inequality (6.1) may be replaced by $6 - 2\sqrt{6}$. In particular, $2\sqrt{3}/3$ is not optimal for that estimate.*

Proof. The Hilbert-space block E in the source example satisfies von Neumann’s inequality

$$\|f(E)\|_2 \leq \sup_{\mathbb{D}} |f|.$$

By Lemma 1,

$$\|f(\tfrac{1}{2}L)\|_p \leq (6 - 2\sqrt{6}) \sup_{\mathbb{D}} |f|.$$

Therefore the direct-sum operator $T = E \oplus \tfrac{1}{2}L$, with the max operator norm used in the source paper, satisfies

$$\|f(T)\| = \max\{\|f(E)\|_2, \|f(\tfrac{1}{2}L)\|_p\} \leq (6 - 2\sqrt{6}) \sup_{\mathbb{D}} |f|,$$

because $6 - 2\sqrt{6} > 1$. Finally,

$$6 - 2\sqrt{6} \approx 1.1010205 < 1.1547005 \approx \frac{2\sqrt{3}}{3}.$$

Thus the displayed constant $2\sqrt{3}/3$ is not optimal. □

Sharpness For The Shift Component

For $p = 1$ or $p = \infty$, the estimate in Lemma 1 is sharp for the shift component. Indeed, for $p = 1$ the matrix of $f(\frac{1}{2}L)$ has column sums tending to $\sum_{k=0}^m |a_k|2^{-k}$, and for $p = \infty$ it has row sum equal to the same expression. Hence

$$\|f(\tfrac{1}{2}L)\|_p = \sum_{k=0}^m |a_k|2^{-k} \quad (p = 1, \infty).$$

The Bohr–Bombieri formula is sharp and its extremals are obtained by disk automorphisms, approximable by polynomials because the relevant pole lies outside $\overline{\mathbb{D}}$. Thus the constant $6 - 2\sqrt{6}$ cannot be lowered for the ℓ^1 or ℓ^∞ shift block.

Scope And Limitations

This packet answers only the optimality of the constant appearing in (6.1). It does not produce a norm-one operator that is polynomially bounded with constant 1 and has infinite numerical-range spectral constant. The full constant-1 question remains open here.

Novelty And Search Notes

Local duplicate check: the run registry, solution index, attempt index, and proof-gap index were searched for arXiv:2410.10678, algebraic numerical range, Bohr–Bombieri, polynomially bounded constant 1, and the constant $2\sqrt{3}/3$. No existing packet for this target or improvement was found.

External/source check: arXiv:2410.10678 was inspected directly from its TeX source and PDF. The supporting Bohr–Bombieri formula was checked in a 2025 paper that explicitly cites Bombieri–Ricci’s 1962 formula. This is recorded as a literature-implied answer rather than an original new proof of the Bohr–Bombieri theorem.

Human Review Recommendation

Reviewers should check only the import of the Bohr–Bombieri formula and the standard computation of the ℓ^1/ℓ^∞ shift norm. Once those are accepted, the replacement of $2\sqrt{3}/3$ by $6 - 2\sqrt{6}$ is immediate.

References

- [1] H. Blazhko, D. Homza, F. L. Schwenninger, J. de Vries, and M. Wojtylak, *The algebraic numerical range as a spectral set in Banach algebras*, arXiv:2410.10678.
- [2] E. Bombieri and G. Ricci, *Sopra un teorema di H. Bohr e G. Ricci sulle funzioni maggioranti delle serie di potenze*, Boll. Un. Mat. Ital. **17**(3), 276–282, 1962.
- [3] R. S. Khasyanov, *Area functional and majorant series estimates in the class of bounded functions in the disk*, arXiv:2503.16313, Section 4.